

Worked Answer Key

For a mean, add all values and divide by the number of values. For a median, order the values first. Range is greatest minus least.

Page 1

1. Ordered 4, 6, 6, 8, 11; sum 35; mean 7; median 6; mode 6; range $11 - 4 = 7$.
2. Ordered 3, 5, 7, 7, 8, 12; sum 42; mean 7; median $(7 + 7) \div 2 = 7$; mode 7; range $12 - 3 = 9$.

Page 2

1. $(12 + 16 + 20 + 24) \div 4 = 72 \div 4 = \mathbf{18}$.
2. $(7 + 9 + 10 + 12 + 17) \div 5 = 55 \div 5 = \mathbf{11 \text{ cans}}$.

Page 3

1. $175 \div 5 = \mathbf{35}$.
2. $96 \div 4 = \mathbf{24 \text{ points}}$.

Page 4

1. Total is $15 \times 4 = 60$. Missing value = $60 - (11 + 14 + 18) = 60 - 43 = \mathbf{17}$.
2. Total is $12 \times 5 = 60$. Missing value = $60 - (8 + 10 + 13 + 15) = 60 - 46 = \mathbf{14}$.

Page 5

1. Ordered 3, 7, 9, 11, 14; median **9**.
2. Ordered 16, 18, 19, 20, 21, 24, 25; median **20**.

Page 6

1. Ordered 4, 6, 8, 10, 12, 14; middle values 8 and 10; median $(8 + 10) \div 2 = \mathbf{9}$.
2. Ordered 18, 20, 22, 25, 27, 29, 31, 35; median $(25 + 27) \div 2 = \mathbf{26}$.

Page 7

1. Ordered 61, 62, 65, 68, 70; median **65°F**.
2. Ordered 85, 95, 110, 120, 140; median **110 pages**.

Page 8

1. 5 occurs three times; mode **5**.
2. 14 occurs four times; mode **14**.

Page 9

1. No mode: every value occurs once.
2. **2 and 4** are modes; each occurs twice.

Page 10

1. $21 - 9 = \mathbf{12}$.
2. Read "fifty-five" as 55: $60 - 45 = \mathbf{15 \text{ seconds}}$.

Page 11

1. Set A range $12 - 4 = 8$. Set B range $14 - 2 = 12$. **Set B** has greater spread.
2. $115 - 98 = \mathbf{17}$.

Page 12

1. $50 \div 5 = 10$. The mean is the equal-share value for the five data points.
2. Ordered 2, 3, 4, 20, 21; median **4**. Median is useful because the two large values pull the mean upward.

Page 13

1. Sum 49; mean $49 \div 6 = 8 \frac{1}{6}$ (about 8.17); median $(7 + 8) \div 2 = 7.5$; mode **7**; range $12 - 5 = 7$.
2. Sum 60; mean $60 \div 5 = 12$; median **11**; mode **11**; range $16 - 9 = 7$.

Page 14

1. Data set is 0, 0, 1, 1, 1, 1, 1, 2, 2, 2, 2, 3. Mode **1** (five families).
2. Range $3 - 0 = 3$ **pets**.

Page 15

1. Data set is 1, 1, 1, 2, 2, 3; sum 10; mean $10 \div 6 = 1 \frac{2}{3}$.
2. Middle values are 1 and 2; median $(1 + 2) \div 2 = 1.5$.

Page 16

1. Total is $20 \times 4 = 80$. Missing value = $80 - (16 + 19 + 25) = 80 - 60 = 20$.
2. Greatest = least + range = $7 + 18 = 25$.

Page 17

1. Original mean = $28 \div 4 = 7$. New mean = $(28 + 12) \div 5 = 40 \div 5 = 8$; it increases by 1.
2. Original ordered set 3, 5, 5, 7, 9 has median 5 and range 6. New ordered set 3, 5, 5, 7, 9, 20 has median $(5 + 7) \div 2 = 6$, range $20 - 3 = 17$.

Page 18

1. Sum 100; mean $100 \div 5 = 20$ **pages**; range $25 - 15 = 10$ **pages**.
2. Ordered 6, 8, 9, 11, 11; median **9 cm**; mode **11 cm**.

Page 19

1. Ordered 18, 18, 18, 21, 24, 27, 30; sum 156; mean $156 \div 7 = 22 \frac{2}{7}$ (about 22.29); median **21**; mode **18**; range $30 - 18 = 12$.
2. Greatest = least + range = $5 + 9 = 14$. (The other stated measures describe the rest of the set; range alone determines this greatest value.)

Page 20

1. Sum 45; mean $45 \div 5 = 9$. Ordered 6, 8, 8, 10, 13; median **8**; mode **8**; range $13 - 6 = 7$.
2. One valid set is **6, 8, 8, 9, 11, 18**. Sum 60, so mean 10; middle values 9 and 11 give median 10; 8 occurs most often, so mode 8; range $18 - 6 = 12$.